

Assignment-1

Reconnaissance Report on a Public Organization

Target Organization: Mozilla

Main Domain: mozilla.org

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1. Introduction and Scope

This report presents a basic reconnaissance activity performed from the viewpoint of a junior security analyst. The selected public organization is Mozilla, and the main domain used for reconnaissance is mozilla.org. The purpose of this assignment is not to attack the organization, but to collect public information that is useful in the early information-gathering phase of a security assessment.

The reconnaissance was performed using passive and safe methods such as public domain lookup, DNS lookup, Google search operators, official organization pages, and technology fingerprinting from Wappalyzer. The Nmap scanning requirement was performed only on scanme.nmap.org, which is the official Nmap test target allowed for learning and testing Nmap commands.

All findings in this report are based on information that was publicly accessible at the time of collection. No unauthorized login, exploitation, brute force attempt, or private data collection was performed.

Selected Targets

Purpose	Target	Reason
Organization reconnaissance	Mozilla / mozilla.org	Public, well-known organization with official leadership and governance pages.
Nmap scanning	scanme.nmap.org	Official Nmap test host for learning and command verification.

2. Methodology and Tools

The assignment was completed using a simple and ethical reconnaissance workflow. First, the public identity of the domain was checked. Second, Google dorking was used to locate visible subdomains and public pages. Third, Wappalyzer was used to identify web technologies. Fourth, Mozilla official pages were used to collect public leadership information and organizational structure. Finally, the required Nmap scans were run on scanme.nmap.org from Windows Command Prompt.

Task	Tool / Source	Purpose
Domain information	ICANN Lookup and nslookup	Collect registration and DNS information.
Subdomain discovery	Google dork: site:*.mozilla.org	Find publicly indexed Mozilla subdomains.
Technology stack	Wappalyzer	Identify visible technologies and web frameworks.
Employee/public information	Mozilla official leadership page	Collect names and roles that are officially public.
Organogram	Mozilla leadership and governance pages	Build a simple public organizational structure.
Nmap scans	Nmap 7.99 on Windows CMD	Perform host discovery, port scan, version detection, and OS detection on scanme.nmap.org.

3. Domain Information

Domain information gives the basic identity of a website. It helps a security analyst understand the registered domain name, DNS configuration, domain status, and public address records associated with the target.

Field	Observed Information
Target domain	mozilla.org
Domain status	clientDeleteProhibited, clientTransferProhibited, clientUpdateProhibited
Nameservers	ns1-240.akam.net, ns4-64.akam.net, ns5-65.akam.net, ns7-66.akam.net
Created	1998-01-24 05:00:00 UTC
Updated	2024-12-27 10:42:44 UTC

Registry expiration	2027-01-23 05:00:00 UTC
DNS resolver used in nslookup	dns.google
Resolved IPv4 address	35.190.14.201
Resolved IPv6 address	2600:1901:0:c197::

Proof of Concept

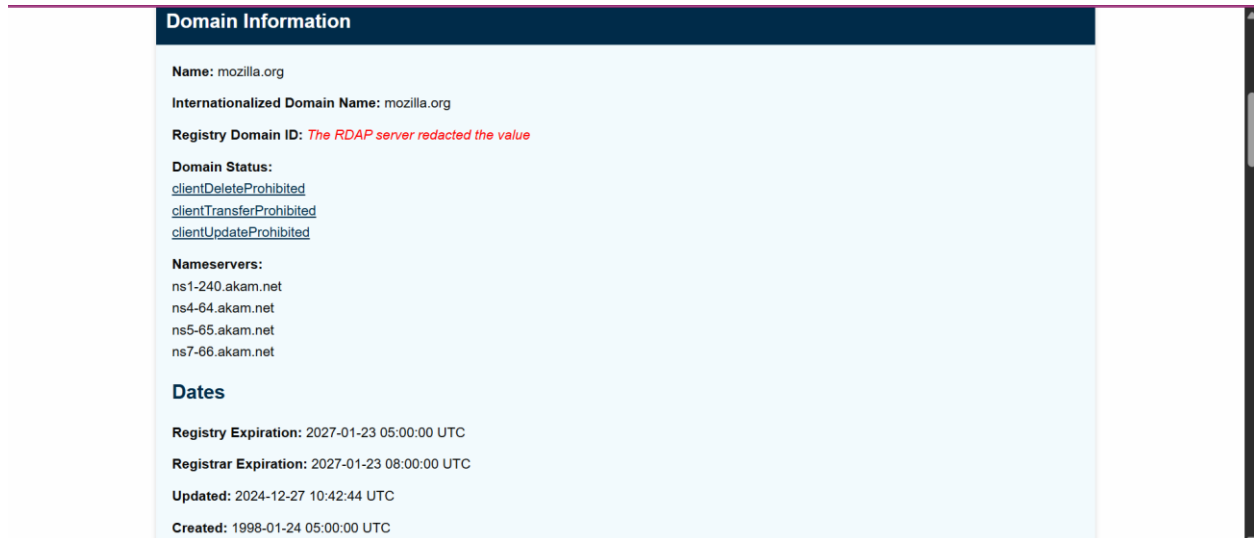


Figure 1: ICANN Lookup result for mozilla.org

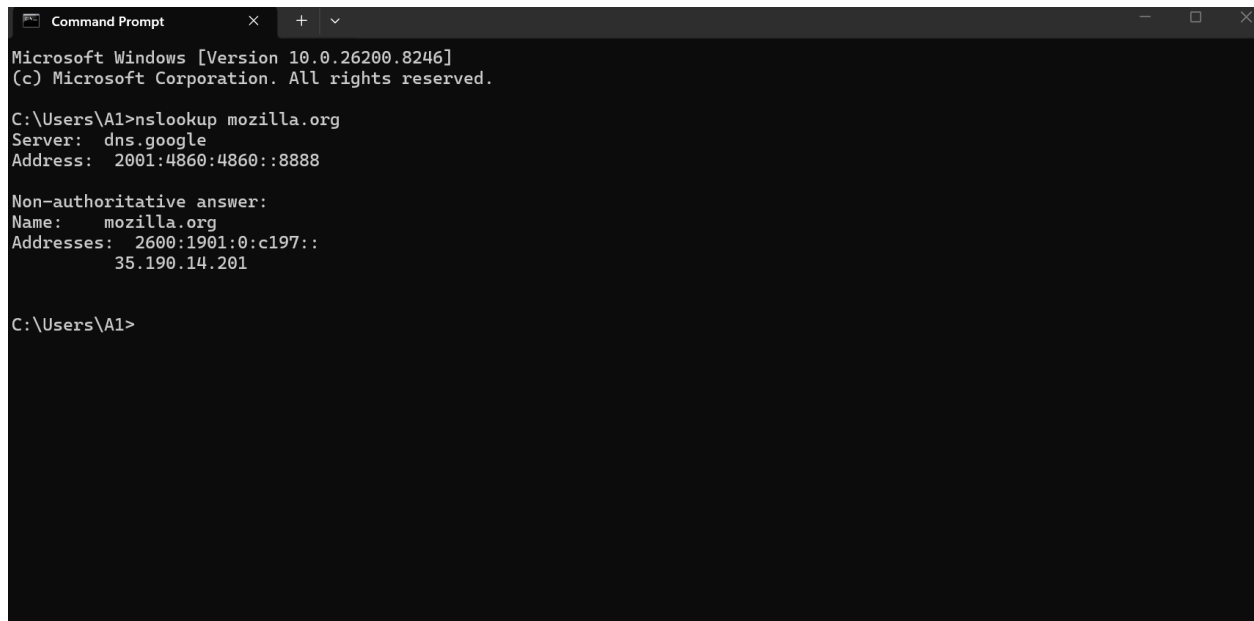


Figure 2: nslookup DNS result for mozilla.org

4. Subdomain Discovery

Subdomains are smaller web areas under a main domain. For example, support.mozilla.org and foundation.mozilla.org are different public services under Mozilla. In this assignment, subdomain discovery was performed passively using Google search operators, so no active scanning was performed against Mozilla.

site:*.mozilla.org

The above Google dork asks Google to show indexed pages from subdomains of mozilla.org. The results showed multiple Mozilla-related subdomains.

Subdomain Found	Observation
connect.mozilla.org	Mozilla Connect community/product feedback page.
builders.mozilla.org	Mozilla Builders project/public initiative page.
stateof.mozilla.org	State of Mozilla public information page.

foundation.mozilla.org	Mozilla Foundation public website.
blog.mozilla.org	Mozilla Blog public page shown in search results.

Proof of Concept

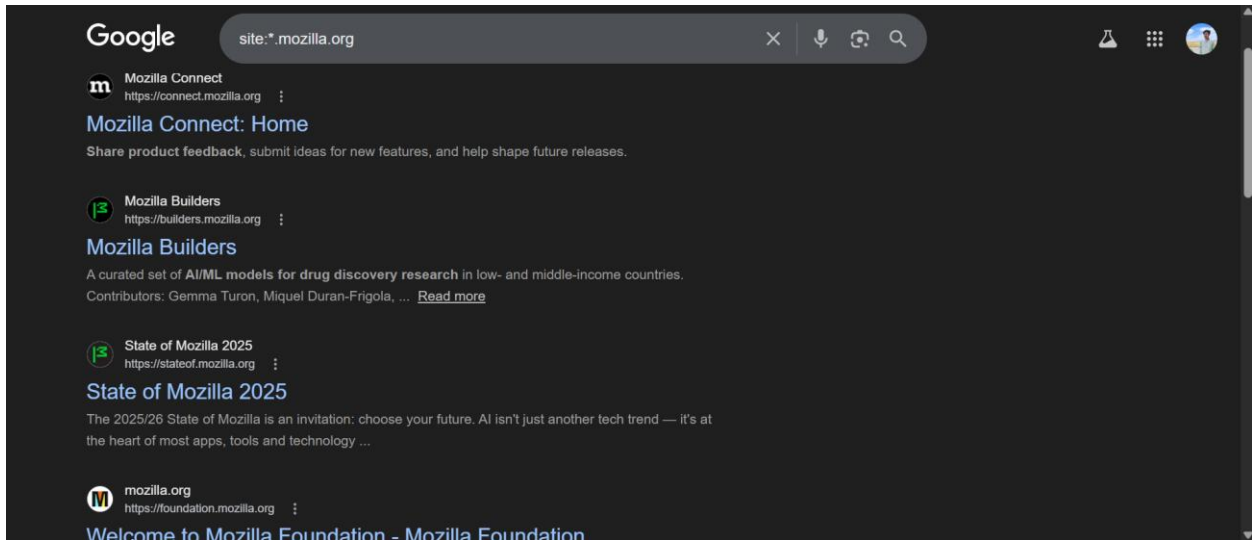


Figure 3: Google dork result showing Mozilla subdomains

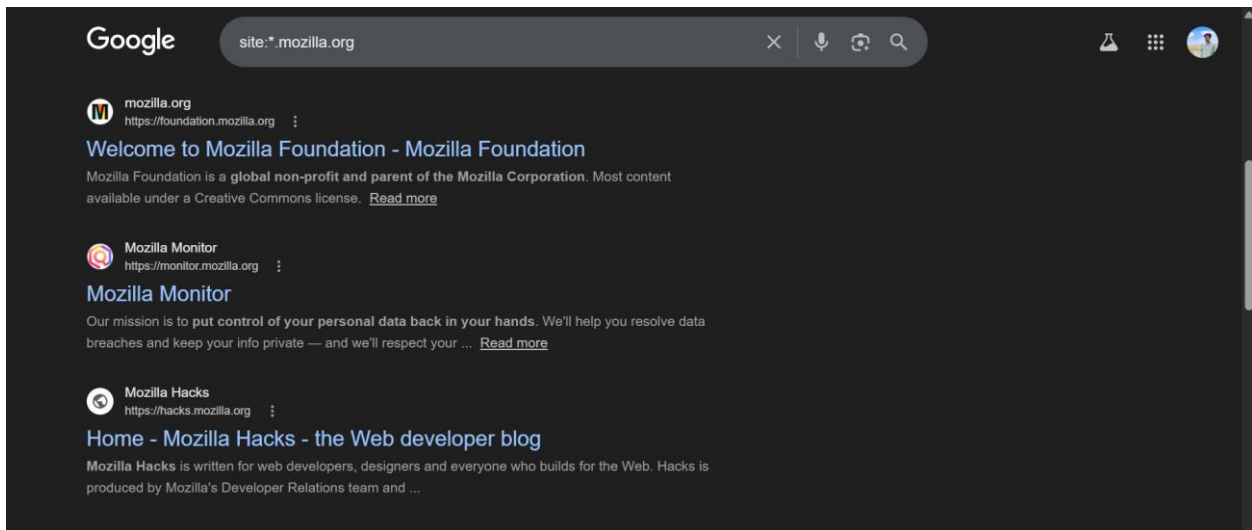


Figure 4: Additional Google dork result for Mozilla-related pages

5. Technology Stack

Technology stack means the set of software, frameworks, services, and tools used to build and run a website. This information helps analysts understand the environment and possible areas for security review.

Category	Detected Technology	Meaning
CMS	Wagtail	A content management system often used with Python/Django websites.
Programming language	Python	Main backend language indicated by Wappalyzer.
Web framework	Django	Python-based web framework.
Caching	Varnish	Caching layer used to improve website performance.
Issue tracking / monitoring	Sentry	Used for error monitoring and application issue tracking.
Development / frontend	Emotion	JavaScript styling library.
Tag manager	Google Tag Manager	Used to manage analytics/marketing scripts.
Email service	Google Workspace	Business email/workspace service.
Payment processor	Stripe	Online payment processing service.

Proof of Concept

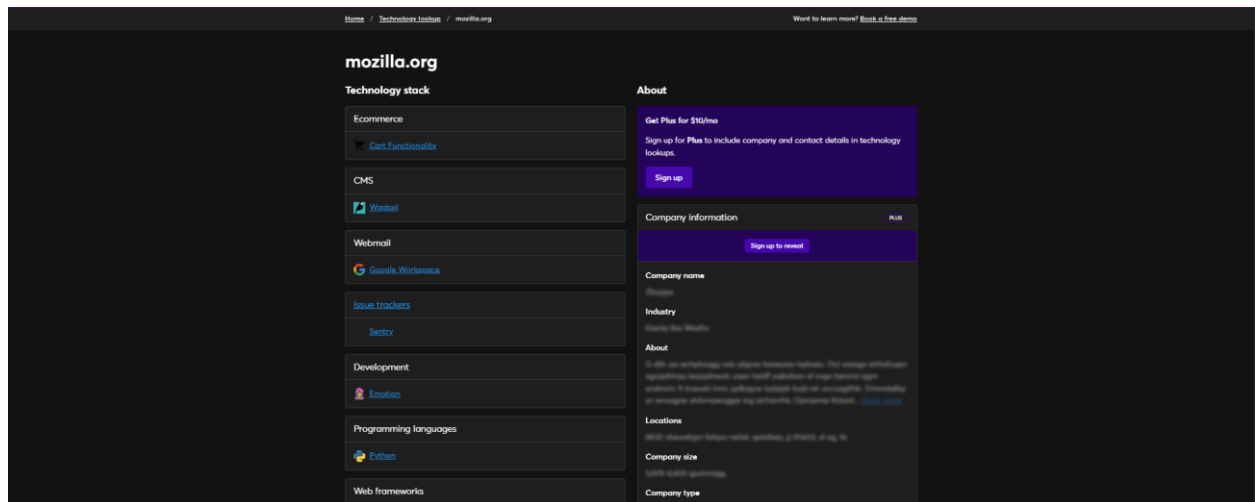


Figure 5: Wappalyzer overview for mozilla.org

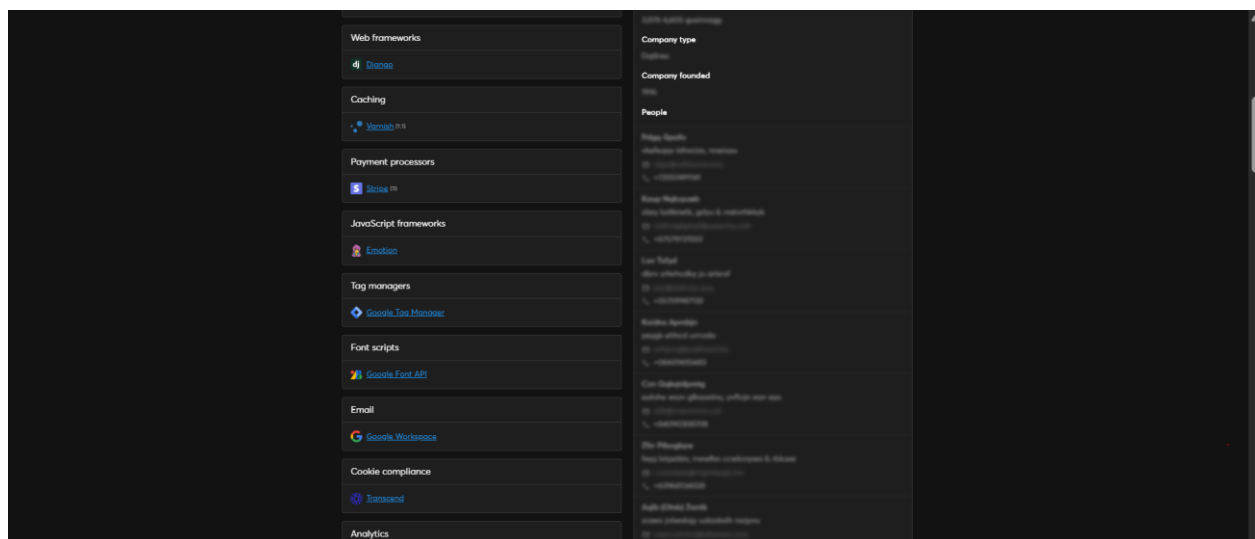


Figure 6: Wappalyzer technology details for mozilla.org

6. Programming Languages

The Wappalyzer result indicates Python as a detected programming language and Django as a web framework. Since this is a public external check, it should be treated as a technology fingerprint rather than full internal confirmation. Like most public websites, browser-side technologies such as HTML, CSS, and JavaScript are also visible to users.

Language / Web Technology	Where Used	Evidence / Reason
Python	Backend / application layer	Detected by Wappalyzer.
Django	Python web framework	Detected by Wappalyzer under web frameworks.
JavaScript	Client-side website behavior	Visible in modern public web pages and supported by detected JS-related technologies.
HTML/CSS	Page structure and styling	Basic client-side technologies required for public webpages.

7. Employee List and Their Public Information

This section uses only official public leadership information. It does not include private employee details, personal contact information, or non-public data. In a reconnaissance report, it is safer and more professional to list public leadership roles rather than trying to collect private information about ordinary staff members.

Name	Public Role	Public Source
Mark Surman	President, Mozilla	Mozilla Leadership page
John Dickerson	CEO, Mozilla.ai	Mozilla Leadership page
Anthony Enzor-DeMeo	Chief Executive Officer, Mozilla Corporation	Mozilla Leadership page
Mohamed Nanabhay	Managing Partner, Mozilla Ventures	Mozilla Leadership page
Ryan Sipes	Managing Director, Product MZLA	Mozilla Leadership page
Nabiha Syed	Executive Director, Mozilla Foundation	Mozilla Leadership page
EM Lewis-Jong	Vice President, Mozilla Data Collective	Mozilla Leadership page
Jenny Choo	VP & Chief of Staff	Mozilla Leadership page
Abigail Besdin	Chief Operating Officer	Mozilla Leadership page
Eric Muhlheim	Chief Financial Officer	Mozilla Leadership page
Stephanie Domas	VP of Security	Mozilla Leadership page
Ajit Varma	Head of Firefox	Mozilla Leadership page

Proof of Concept

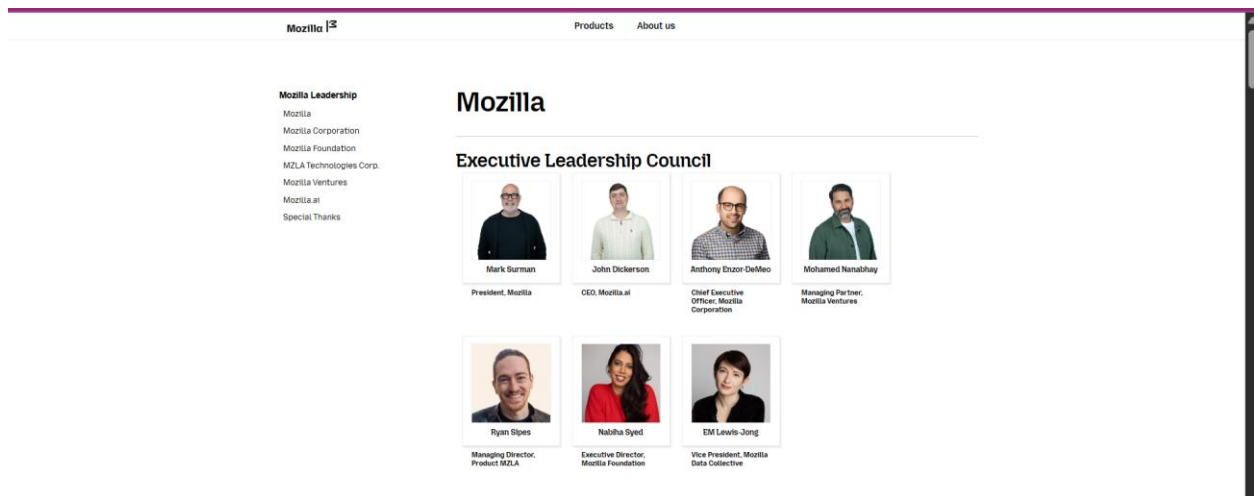
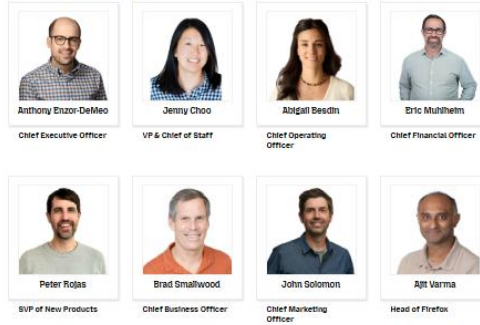


Figure 7: Mozilla Executive Leadership Council from official page

Mozilla Corporation

Executive Team



Senior Leadership Team

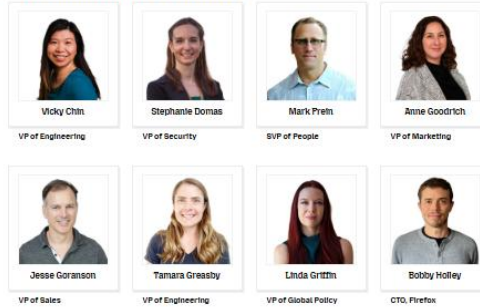


Figure 8: Mozilla Corporation executive and senior leadership

Mozilla Foundation

Leadership Team



Board of Directors

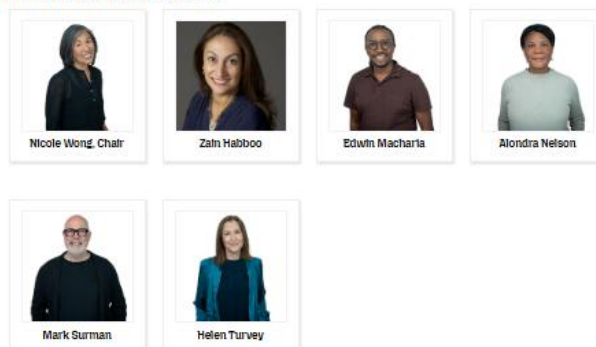


Figure 9: Mozilla Foundation leadership and board section

8. Organizational Structure / Organogram

An organogram shows the structure of an organization. Based on Mozilla official leadership and governance pages, Mozilla can be represented using a simple public structure. This is not a private internal chart; it is a public-facing structure prepared from available information.

```

Mozilla
|-- Executive Leadership Council
|-- Mozilla Corporation
|   |-- Executive Team
|   |-- Senior Leadership Team
|-- Mozilla Foundation
|   |-- Leadership Team
|   |-- Board of Directors
|-- MZLA Technologies Corp.
|-- Mozilla Ventures
|-- Mozilla.ai
|-- Mozilla Data Collective
  
```

Mozilla also describes itself as an open source project with a virtual organization where authority is distributed among volunteer and employed community members based on contribution. This makes Mozilla different from a simple traditional company hierarchy.

Proof of Concept

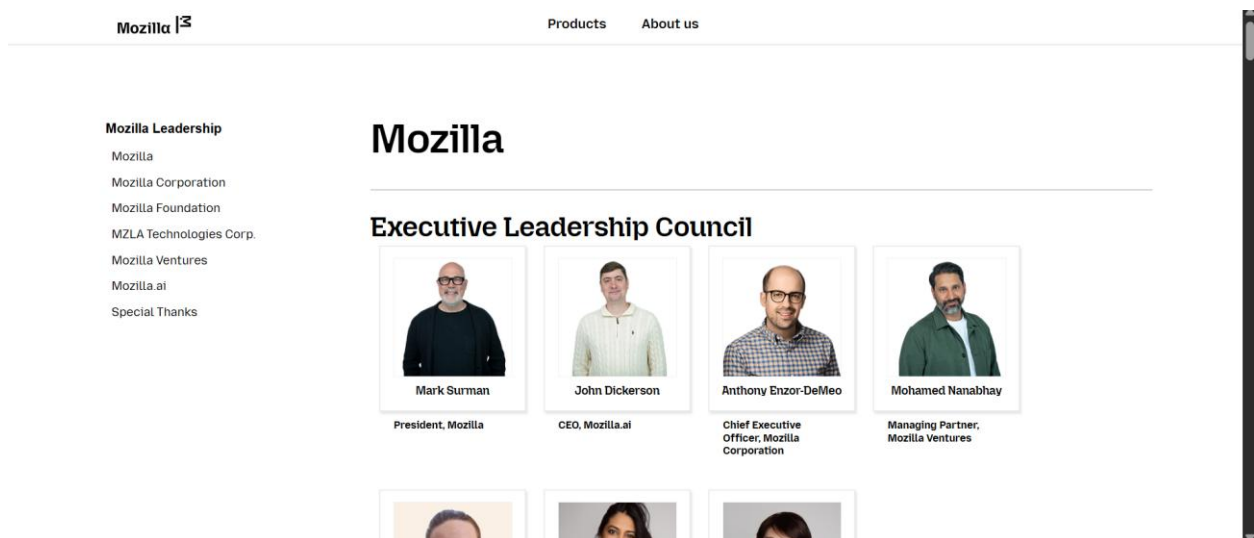


Figure 10: Mozilla leadership page showing public organizational sections

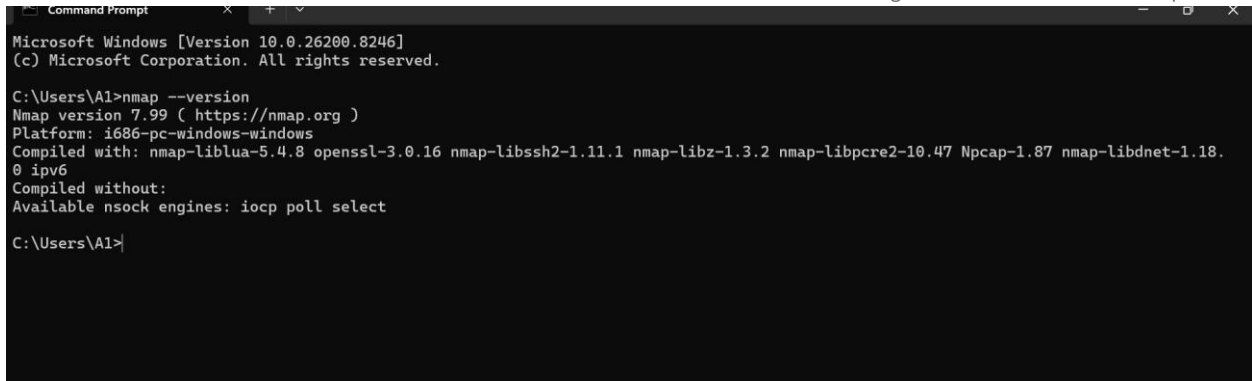
9. Nmap Scanning on scanme.nmap.org

The Nmap portion of the assignment was performed on scanme.nmap.org, not on Mozilla. This follows the assignment requirement and keeps the work safe and legal. Nmap scan results can change over time depending on the target configuration, network conditions, and Nmap version.

9.1 Nmap Version Check

```
nmap --version
```

This command confirms that Nmap was installed and available on the Windows system before scanning.



```

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\A1>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.4 nmap-libnet-1.1.8
0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\A1>

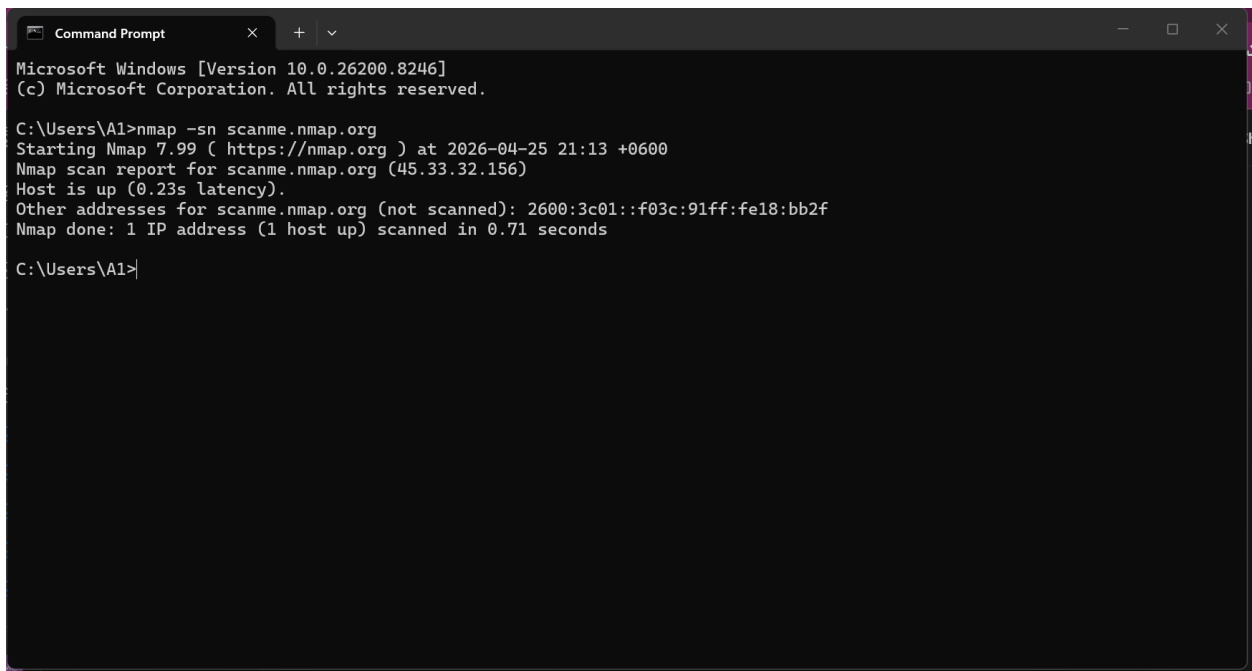
```

Figure 11: Nmap version check on Windows

9.2 Host Discovery

```
nmap -sn scanme.nmap.org
```

Host discovery checks whether the target host is reachable. The result showed that scanme.nmap.org was up and reachable at IP address 45.33.32.156.



```

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\A1>nmap -sn scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-25 21:13 +0600
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 0.71 seconds

C:\Users\A1>

```

Figure 12: Host discovery scan result

9.3 Basic Port Scan - Top 1000 Ports

```
nmap scanme.nmap.org
```

The basic Nmap scan checks the most common 1000 TCP ports. The result showed several open ports and some filtered ports.

Port	State	Service	Meaning
22/tcp	open	ssh	Remote secure shell service.
80/tcp	open	http	Web service.
9929/tcp	open	nping-echo	Nping echo service used for testing.
31337/tcp	open	Elite	Test/open service name shown by Nmap.

```

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\A1>nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-25 21:15 +0600
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
646/tcp    filtered ldap
1719/tcp   filtered h323gatestat
1720/tcp   filtered h323q931
2001/tcp   filtered dc
5060/tcp   filtered sip
5061/tcp   filtered sip-tls
9929/tcp   open  nping-echo
31337/tcp  open  Elite
49152/tcp  filtered unknown

Nmap done: 1 IP address (1 host up) scanned in 7.73 seconds

C:\Users\A1>

```

Figure 13: Basic port scan result for scanme.nmap.org

9.4 Service and Version Detection

```
nmap -sV scanme.nmap.org
```

Service and version detection identifies the application/service running on open ports and attempts to detect version information. This is useful because outdated service versions can indicate possible security risks during a real authorized assessment.

Port	State	Detected Service	Detected Version / Detail
22/tcp	open	tcpwrapped	Service wrapped/limited response.
80/tcp	open	http	Apache httpd 2.4.7 ((Ubuntu))
9929/tcp	open	nping-echo	Nping echo
31337/tcp	open	tcpwrapped	Service wrapped/limited response.

```

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\A1>nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-25 21:18 +0600
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
646/tcp    filtered ldap
1719/tcp   filtered h323gatestat
1720/tcp   filtered h323q931
2001/tcp   filtered dc
5060/tcp   filtered sip
5061/tcp   filtered sip-tls
9929/tcp   open  nping-echo   Nping echo
31337/tcp  open  tcpwrapped
49152/tcp  filtered unknown

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.68 seconds

C:\Users\A1>

```

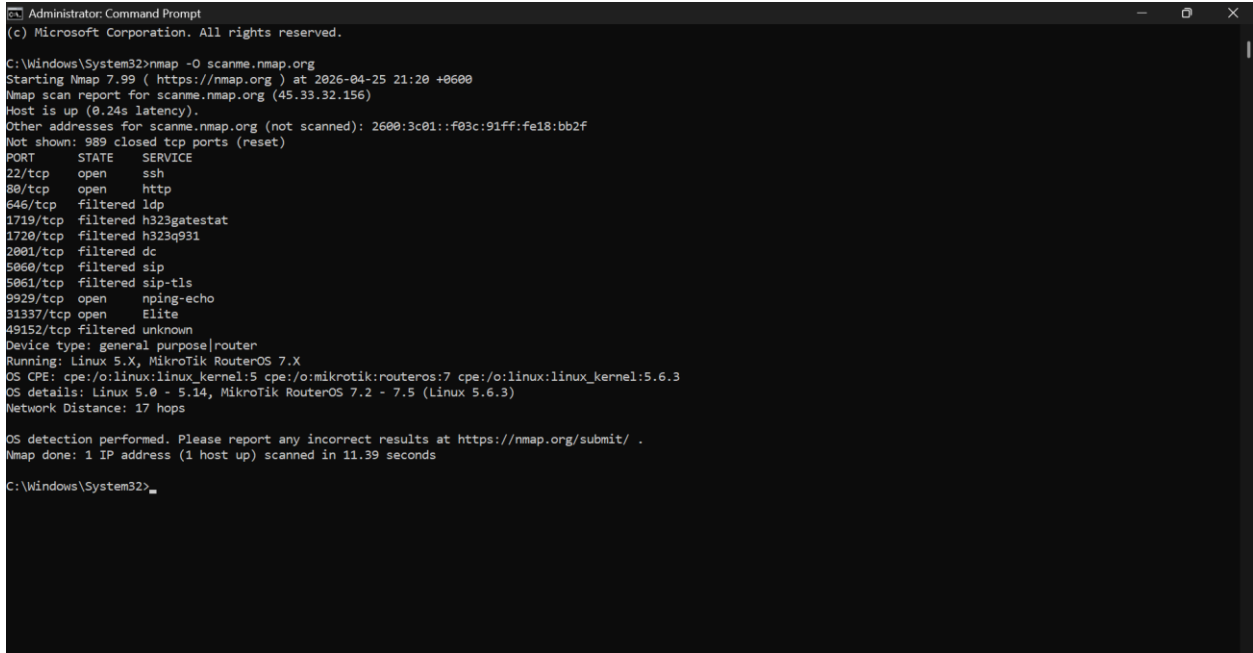
Figure 14: Service and version detection scan result

9.5 OS Detection

```
nmap -O scanme.nmap.org
```

OS detection tries to identify the operating system of the target by analyzing network responses. The result suggested Linux 5.x / MikroTik RouterOS 7.x related fingerprints. OS detection is a best-effort guess and should not be treated as 100% confirmed without additional validation.

Item	Observed Result
Device type	general purpose router
Running	Linux 5.X, MikroTik RouterOS 7.X
OS details	Linux 5.0 - 5.14, MikroTik RouterOS 7.2 - 7.5
Network distance	17 hops



```

Administrator: Command Prompt
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>nmap -O scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-25 21:20 +0600
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
646/tcp   filtered ldp
1719/tcp  filtered h323gatestat
1720/tcp  filtered h323q931
2001/tcp  filtered dc
5060/tcp  filtered sip
5061/tcp  filtered sip-tls
9929/tcp  open  nping-echo
31337/tcp open  Elite
49152/tcp filtered unknown
Device type: general purpose|router
Running: Linux 5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 5.0 - 5.14, MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 17 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.39 seconds

C:\Windows\System32>

```

Figure 15: OS detection result for scanme.nmap.org

10. Conclusion

This assignment demonstrated the early reconnaissance phase of security analysis. The Mozilla domain was reviewed using public sources, Google dorking, DNS lookup, and technology fingerprinting. Public leadership pages were used to prepare the employee/public information and organogram sections. The Nmap section was completed on scanme.nmap.org as required, covering host discovery, basic port scanning, service/version detection, and OS detection.

The main lesson from this assignment is that a large amount of useful security context can be gathered without attacking a target. Ethical reconnaissance focuses on public information, proper permission, and careful documentation of evidence.

Ethical Statement

All reconnaissance activities in this report were performed for academic learning. The Mozilla organization was not attacked or scanned with Nmap. Only public websites and public search results were used for the Mozilla reconnaissance section. The Nmap scans were performed only on scanme.nmap.org, the official Nmap test target. No exploitation, denial-of-service activity, password attack, or unauthorized access was attempted.

11. References

- Mozilla Leadership. <https://www.mozilla.org/en-US/about/leadership/>
- Mozilla Governance. <https://www.mozilla.org/en-US/about/governance/>
- ICANN Lookup. <https://lookup.icann.org/>

- Wappalyzer Technology Lookup. <https://www.wappalyzer.com/lookup/>
- Nmap ScanMe test target. <https://scanme.nmap.org/>
- Nmap Network Scanning - Examples. <https://nmap.org/book/man-examples.html>
- Nmap Legal Issues / ScanMe permission. <https://nmap.org/book/legal-issues.html>